DOMINANCE FROM A SCHEME WITH FINITELY MANY IRREDUCIBLE COMPONENTS IS DETECTED ON GENERIC POINTS OF THE IRREDUCIBLE COMPONENTS OF THE TARGET

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-dominant-finite-number-irreducible-components is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 1382-1390 and proof span 1392-1405 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The hypothesis in the statement is written by the author as '(if and) only if', and that wording is kept verbatim rather than normalised to a plain equivalence.

Allowed standard prerequisites (editorial).

- Irreducible components and generic points: an irreducible closed subset has a unique generic point, and the closure of the image of an irreducible subset under a continuous map is irreducible. Used at proof lines 1395-1396 to pass from the components Z_i of X to the irreducible subsets closure of f(xi_i) in S. Supplies the generic points xi_i and the irreducibility of each f(Z_i).
- A morphism is dominant exactly when the closure of its image is the whole target, and the closure of a finite union is the union of the closures. Used at proof lines 1397-1399 in the display S = closure of f(X) = union of closures of f(Z_i). Turns the hypothesis that f is dominant into a finite union decomposition of S by irreducible subsets.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. Suppose that X has finitely many irreducible components. Then f is dominant (if and) only if for every irreducible component $Z \subset S$ the generic point of Z is in the image of f. If so, then S has finitely many irreducible components as well.*

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Proof. Assume f is dominant. Say $X = Z_1 \cup Z_2 \cup \ldots \cup Z_n$ is the decomposition of X into irreducible components. Let $\xi_i \in Z_i$ be its generic point, so $Z_i = \overline{\{\xi_i\}}$. Note that $f(Z_i)$ is an irreducible subset of S. Hence

$$S = \overline{f(X)} = \bigcup \overline{f(Z_i)} = \bigcup \overline{\{f(\xi_i)\}}$$

is a finite union of irreducible subsets whose generic points are in the image of f. The lemma follows. $\square$